

ses also highlight that model judgments can be confounded by semantic content rather than pure style. The challenge of disentangling an author’s unique style from the topic of the text is well-recognized in authorship analysis (Alshomary et al., 2024).

A.3 Interpretability

Lyu et al. (2023) identified specific latent directions corresponding to concrete stylistic attributes – such as formality and lexical complexity – in a pre-trained model’s embedding space. Their findings provide evidence that certain stylistic features are encoded along approximately linear axes, making them separable by simple geometric probes. Such results echo the broader concept-vector findings above, but for attributes of writing style. Still, uncovering literary style (encompassing a complex mix of diction, syntax, and narrative voice) may pose an even greater challenge than these relatively focused style elements. Interpretability research has begun to bridge these latent representations with human-interpretable descriptions. Some approaches aim to map regions or dimensions of embedding space to understandable concepts. For example, Simhi and Markovitch (2023) project predefined semantic concepts into a model’s embedding space, using them as basis vectors to interpret other embeddings. Another line of work leverages generative LLMs to produce interpretable style representations: Patel et al. (2023) used GPT-3 to annotate millions of sentences with stylistic descriptors and distilled these into a “Linguistically Interpretable Style Embedding” model. The resulting system encodes texts into a 768-dimensional style vector aligned with attributes like formality, tone, and syntax, allowing direct inspection of which dimensions are active for a given text. Together, these efforts underscore both the richness of style information in LLM latent spaces and the complexity of extracting or explaining it.

B Methods

The code for generating and processing data will be made available on a GitHub repository.

B.1 Dataset

In Tab. 3, we present the references to the authors and literary pieces used as the input data to our analysis.

We introduce one text in French (G. Sand’s *Jacques*) as a baseline for separability (embeddings

from different languages are usually easy to differentiate). The other texts include different anglophone authors with distinctive styles yet from about the same time period, and from different English-speaking countries. This is in order to maintain a certain homogeneity in the style. We also considered only rather large novels (over 500 pages) in order to be able to assemble substantial ensembles.

B.2 Excerpts

The full text of a given novel is tokenized at once, and the resulting sequence of token identifiers is divided into chunks of N consecutive tokens. These non-overlapping chunks represent short text fragments of various forms. In particular, they do not necessarily start or end with a sentence and the last token can be anything: punctuation, word suffix, article, verb, etc. In Tab. 4, we show a few examples of 16-token chunks in their textual form.

B.3 Classifiers

For binary classification, we use a Support Vector Machine with linear kernel, and after dimensionality reduction by PCA to 64 dimensions. The training to validation data split was 70/30.

For multiclass classification, we train a Multi-layer Perceptron with penultimate layer of dimension 32, cross-entropy loss and Adam optimizer.

C Map of style

Fig. 5 presents a visualization of style proximity across books and authors. Here we propose an alternative representation and interpret it from a traditional literature analysis perspective.

C.1 Centroid visualization

In order to emphasize proximity between ensembles (rather than text snippets), we propose an alternative representation based on centroid proximity. We calculate the centroids location of each book ensemble and apply multidimensional scaling (MDS) to yield a two-dimensional representation in Fig. 6. The similarity matrix used for MDS is the pair cosine distances between centroids.

C.2 Literary comment

This spatial distribution of Fig. 6 depicts interesting relationships between the narrative techniques and styles of various authors. Virginia Woolf (VW) and James Joyce (JJ), known for pioneering techniques like stream of consciousness

ID	Author	Novel	Date	Country
GS	George Sand	<i>Jacques*</i>	1833	France
HM1	Herman Melville	<i>Moby Dick</i>	1851	America
HM2	Herman Melville	<i>Pierre</i>	1852	America
JA1	Jane Austen	<i>Emma</i>	1815	England
JA2	Jane Austen	<i>Pride and Prejudice</i>	1813	England
JA3	Jane Austen	<i>Sense and Sensibility</i>	1811	England
JJ	James Joyce	<i>Ulysses</i>	1922	Ireland
MT1	Mark Twain	<i>Life on the Mississippi</i>	1883	America
MT2	Mark Twain	<i>Roughing It</i>	1872	America
NH1	Nathaniel Hawthorne	<i>The House of the Seven Gables</i>	1851	America
NH2	Nathaniel Hawthorne	<i>The Scarlet Letter</i>	1850	America
VW1	Virginia Woolf	<i>Night and Day</i>	1919	England
VW2	Virginia Woolf	<i>The Voyage Out</i>	1915	England

Table 3: Authors and novels used for analysis. Note that *Jacques* is in French.

15 preceding tokens	last token
least knew somebody who knew his father and mother? To the peasants of old	time
off time superstition clung easily round every person or thing that was at all	unw
crime; especially if he had any reputation for knowledge, or showed any skill	in
live in a rollicking fashion, and keep a jolly Christmas,	Wh
shook him, and his limbs were stiff, and his hands clutched the	bag
nothing strange for people of average culture and experience, but for the villagers near	whom
road, and lifting more imposing fronts than the rectory, which	peeped
which seemed to explain things otherwise incredible; but the argumentative Mr.	Macey
handicraft. All cleverness, whether in the rapid use of that difficult	instrument
lar or the knife-grinder. No one knew where wandering men had	their
-weaver, named Silas Marner, worked at his vocation in	a
of it, and two or three large brick-and-stone homesteads	,
the outskirts of civilization—inhabited by meagre sheep and thinly-	sc
certain awe at the mysterious action of the loom, by a pleasant sense of	scorn
The questionable sound of Silas's loom, so unlike the natural cheerful	tro

Table 4: Examples of 16-token excerpts from G. Eliot's *Silas Marner*. Embeddings derived from the last token are the ones collected to form the ensembles of the analysis.

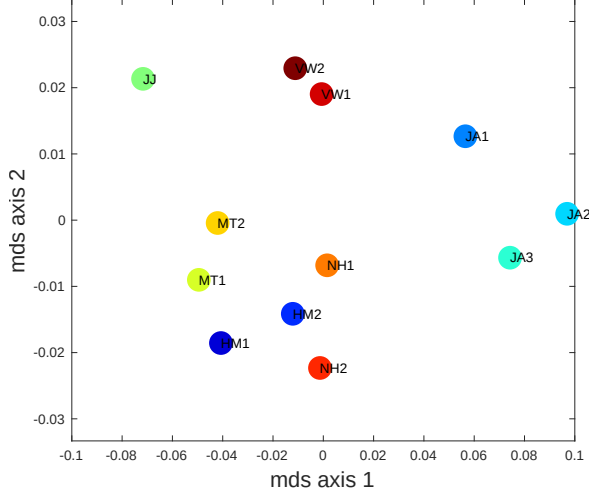


Figure 6: Alternative map of style emphasizing relative distances between ensembles by embedding their shared geometry with multidimensional scaling.

and free indirect discourse, cluster apart, reflecting their shared modernist experimentation. Jane Austen’s texts (JA), which also employ free indirect discourse but within a more traditional realist framework, form their own distinct grouping. Austen’s group is clearly separated both from the modernists and from American authors such as Nathaniel Hawthorne (NH), Herman Melville (HM), and Mark Twain (MT). These American writers, characterized predominantly by narrative realism or romanticism, are grouped centrally and distinctly apart from the experimental modernist approaches of Woolf and Joyce.

D Generalization to additional models

For generalization, we reproduce the same methodology with three other open-source models released in 2025:

- gemma-3-1b-pt from Google (US) (Team et al., 2025)
- Qwen3-1.7B-Base from Qwen (China) (Yang et al., 2025)
- SmoLLM2-1.7B from Hugging Face (France) (Allal et al., 2025)

These models are queried in their “base” form, i.e. not fine-tuned for chat (“instruct”). We use models in the one-billion-parameter range as smaller models are generally more interpretable, and also due to compute constraints.

We find the same patterns of ensemble separability across models, as shown in Fig. 7. In particu-

lar, embeddings corresponding to increased context and deep layers are more easily separable.

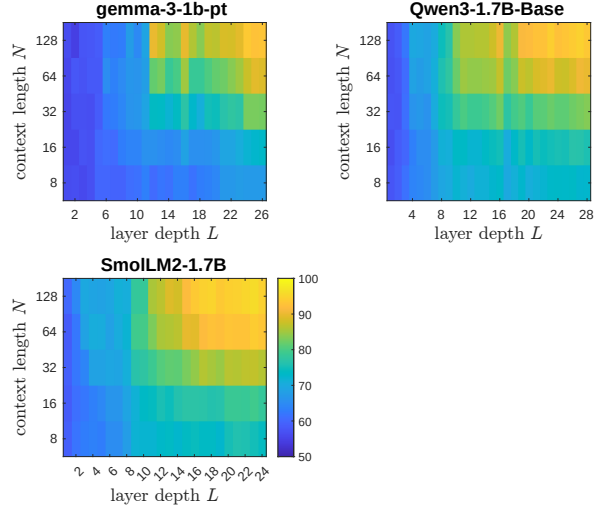


Figure 7: Probe accuracy (%) for GE vs VW in other LLMs (applied on deepest embeddings).